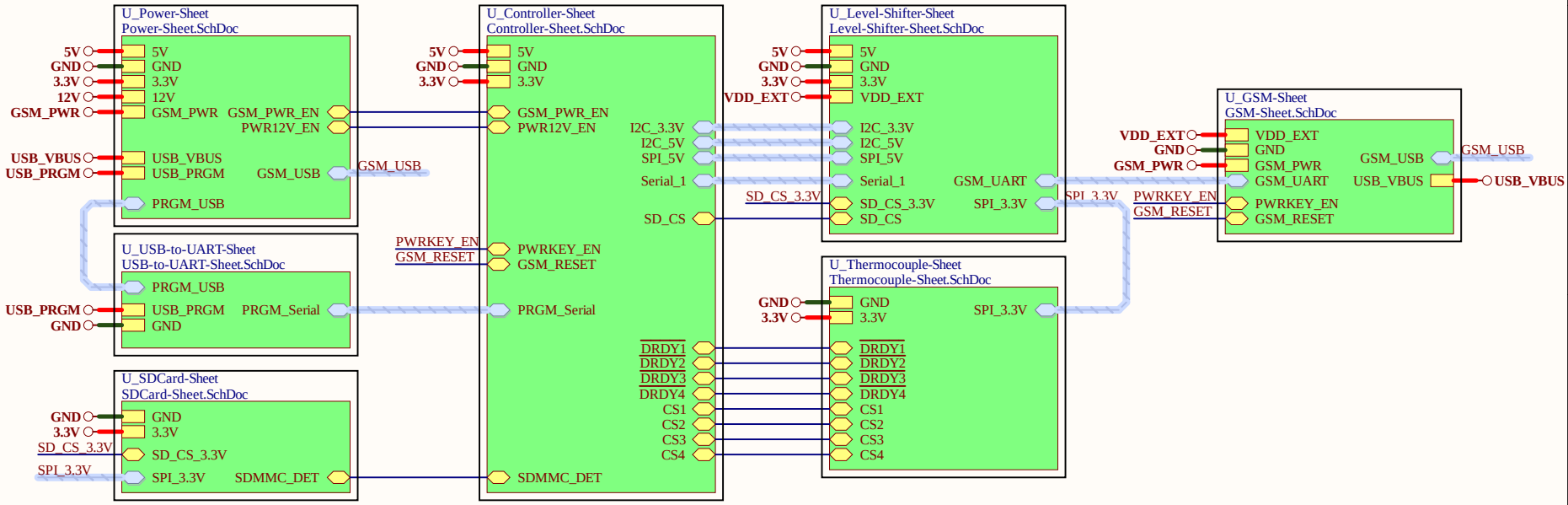
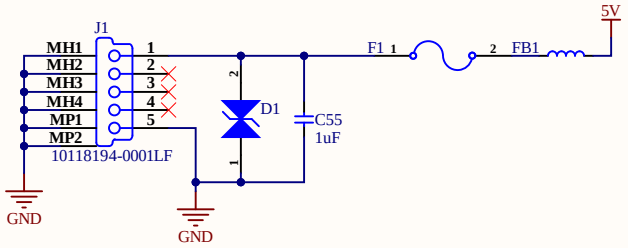


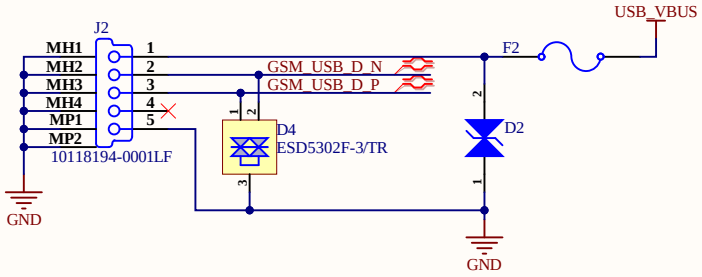
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Drawn By: Huzaifa Ali	Page Total: 8

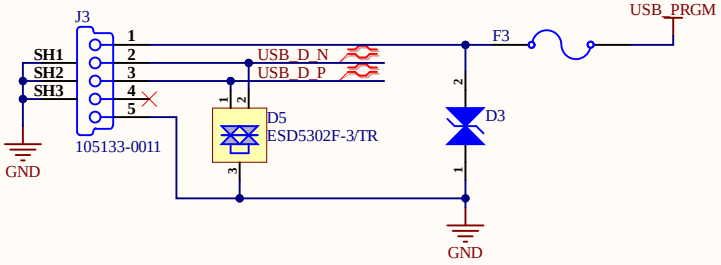
USB FOR BOARD POWER



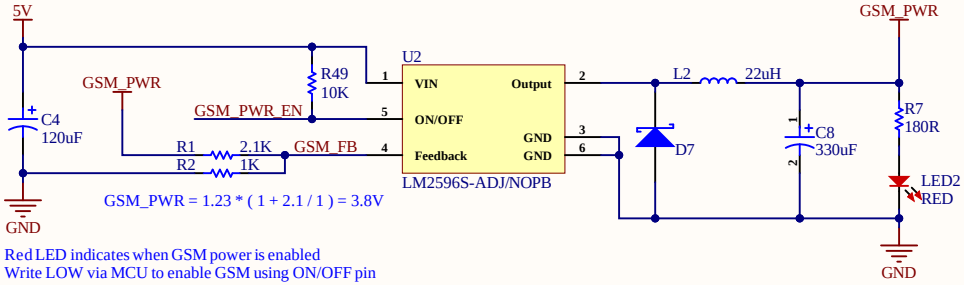
USB FOR GSM UPDATES



USB PROGRAMMING MEGA2560

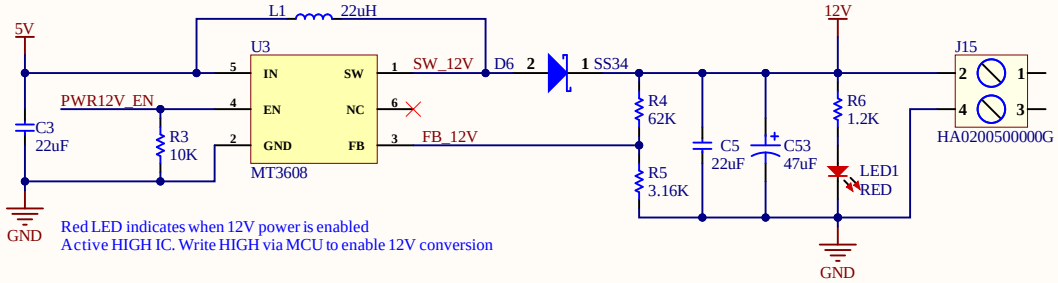


BUCK FOR GSM POWER 5V TO 3.8V



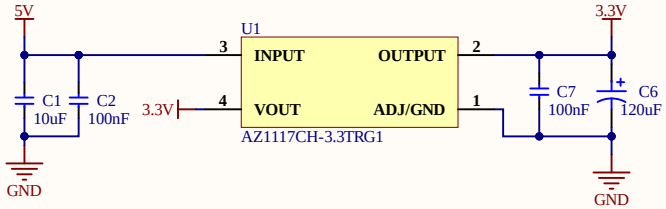
Red LED indicates when GSM power is enabled
Write LOW via MCU to enable GSM using ON/OFF pin

BOOST FOR WIKA SENSOR 5V TO 12V



Red LED indicates when 12V power is enabled
Active HIGH IC. Write HIGH via MCU to enable 12V conversion

LDO 5V TO 3.3V



Test Points

P1 — 5V
P2 — USB_VBUS
P3 — USB_PRGM
P4 — GSM_PWR
P5 — 12V
P6 — 3.3V

GSM_USB ⇄ GSM_USB_D_N, GSM_USB_D_P

PRGM_USB ⇄ USB_D_N, USB_D_P

GSM_PWR_EN ⇄ PWR12V_EN

USB_VBUS	USB_VBUS
USB_PRGM	USB_PRGM
GSM_PWR	GSM_PWR
12V	12V
5V	5V
3.3V	3.3V
GND	GND

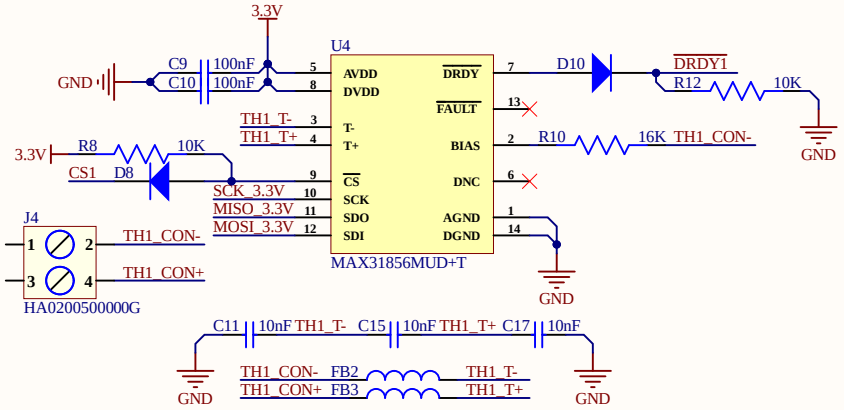
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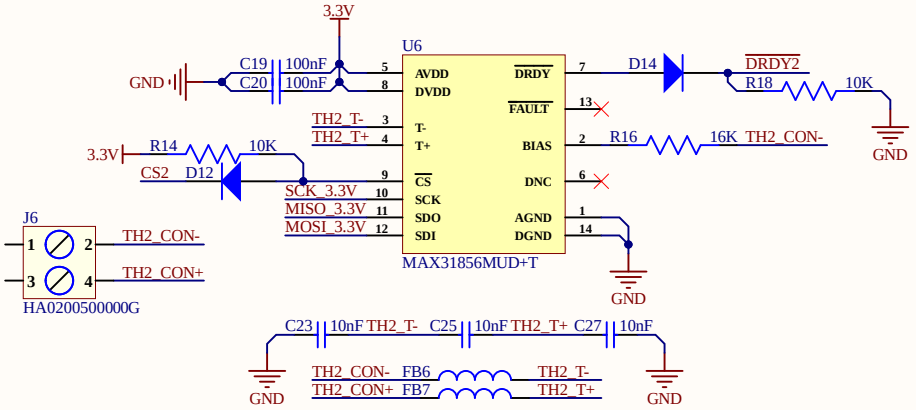
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Date: 5/11/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

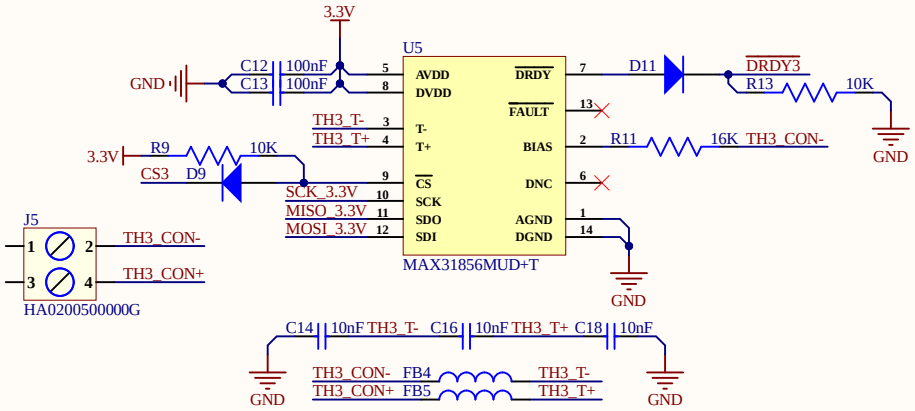
Thermocouple Input - 1



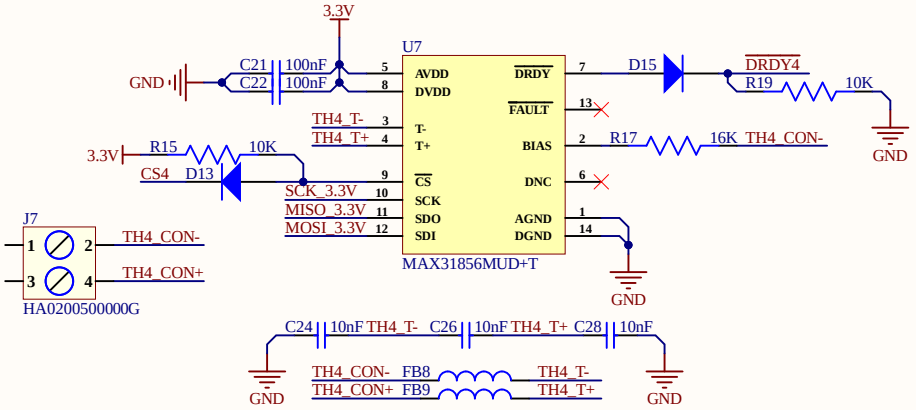
Thermocouple Input - 2



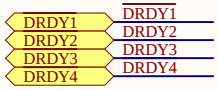
Thermocouple Input - 3

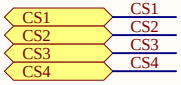


Thermocouple Input - 4







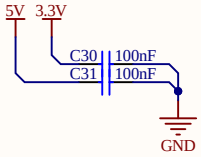


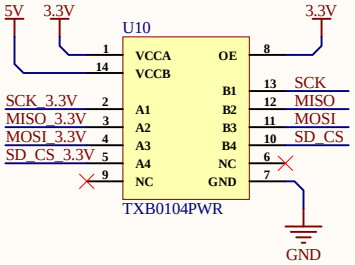


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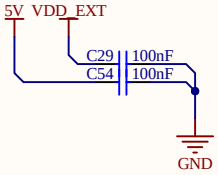
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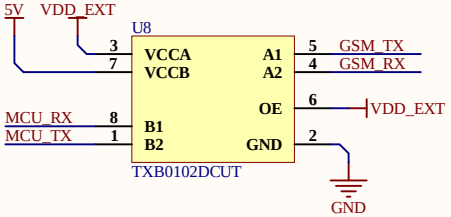
Level Shifter for SPI



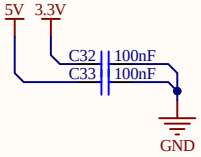


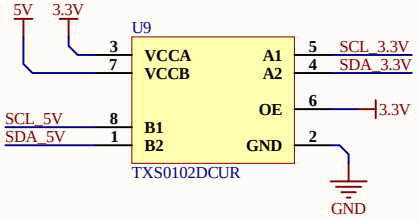
Level Shifter for Serial

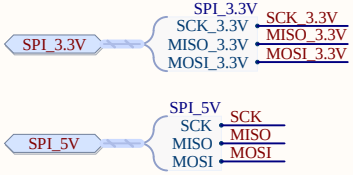


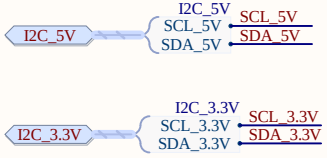


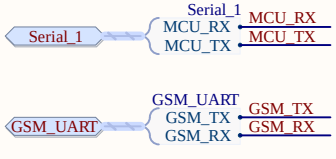
Level Shifter for I2C

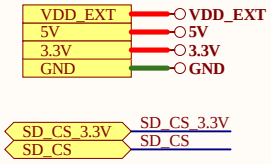






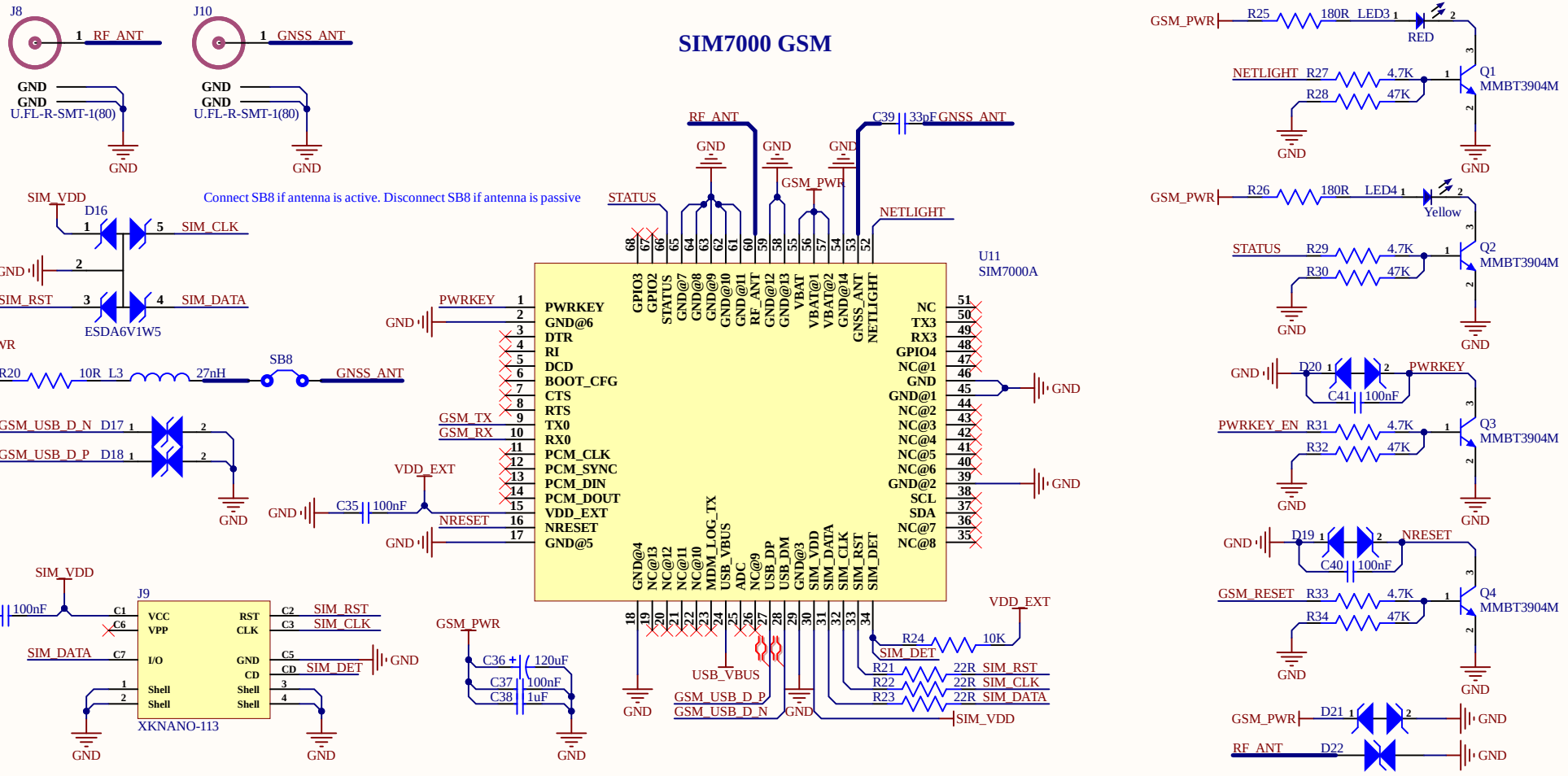


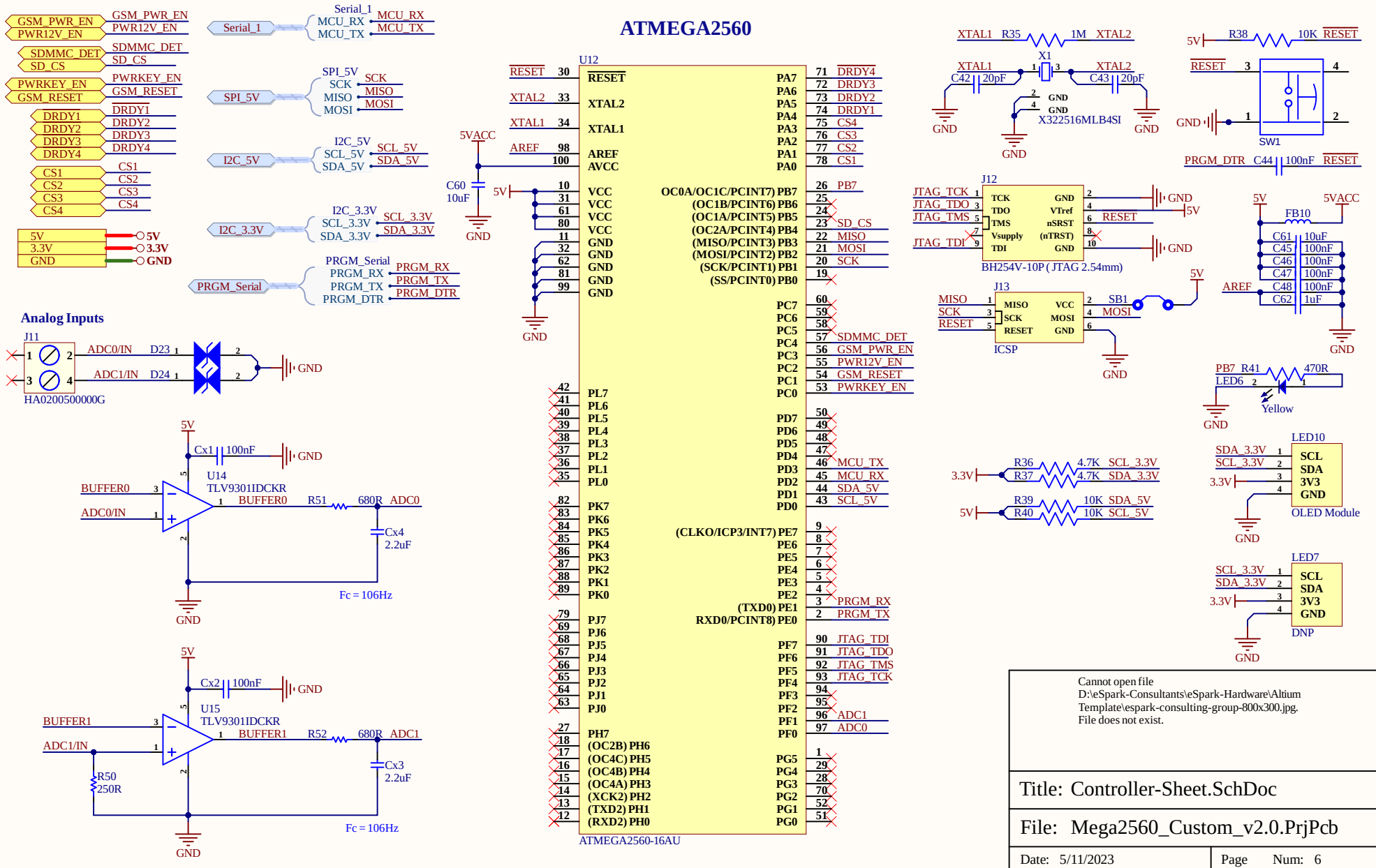




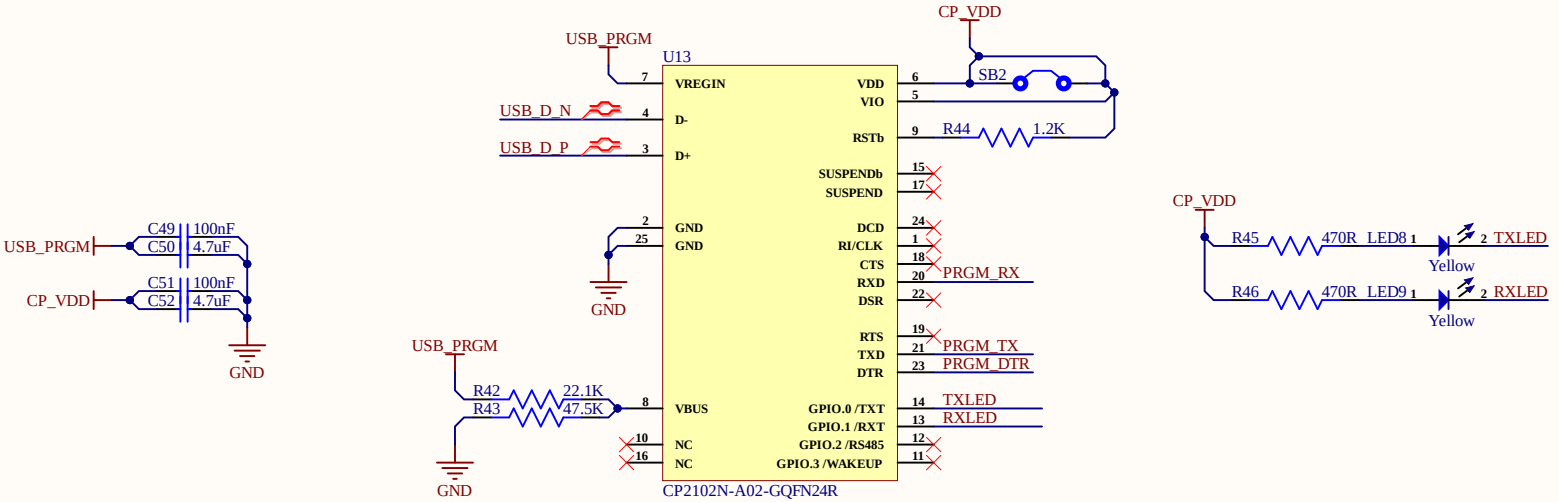
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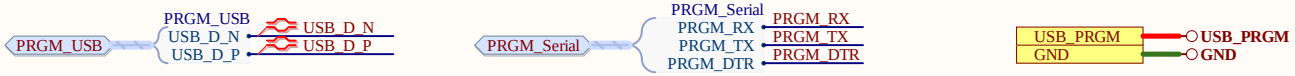
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Date: 5/11/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8


SIM7000 GSM
J8
J10
RF_ANT
GNSS_ANT
U.FL-R-SMT-1(80)
Connect SB8 if antenna is active. Disconnect SB8 if antenna is passive
SIM_VDD
D16
ESDA6V1W5
SIM_CLK
SIM_DATA
SIM_RST
GSM_PWR
R20
10R
L3
27nH
SB8
GSM_USB_D_N
GSM_USB_D_P
D17
D18
C35
100nF
VDD_EXT
NRESET
PWRKEY
GSM_TX
GSM_RX
STATUS
NETLIGHT
C39
33pF
U11
SIM7000A
PWRKEY
GND@6
DTR
RI
DCD
BOOT_CFG
CTS
RTS
TX0
RX0
PCM_CLK
PCM_SYNC
PCM_DIN
PCM_DOUT
VDD_EXT
NRESET
GND@5
GPIO3
GPIO2
STATUS
GND@7
GND@8
GND@9
GND@10
GND@11
RF_ANT
GND@12
GND@13
VBAT
VBAT@1
VBAT@2
GND@14
GNSS_ANT
NETLIGHT
NC
TX3
RX3
GPIO4
NC@1
GND
GND@1
NC@2
NC@3
NC@4
NC@5
NC@6
GND@2
SCL
SDA
NC@7
NC@8
GND@4
NC@13
NC@12
NC@11
NC@10
MDM_LOG_TX
USB_VBUS
ADC
NC@9
USB_DP
USB_DM
GND@3
SIM_VDD
SIM_DATA
SIM_CLK
SIM_RST
SIM_DET
J9
XKNANO-113
C34
100nF
C36
120uF
C37
100nF
C38
1uF
R24
10K
R21
22R
R22
22R
R23
22R
SIM_RST
SIM_CLK
SIM_DATA
SIM_DET
R25
180R
LED3
RED
R26
LED4
Yellow
R27
4.7K
R28
47K
Q1
MMBT3904M
R29
R30
Q2
C41
D20
PWRKEY_EN
R31
R32
Q3
C40
D19
NRESET
GSM_RESET
R33
R34
Q4
D21
D22
GSM_UART
GSM_TX
GSM_RX
GSM_USB
GSM_USB_D_N
GSM_USB_D_P
PWRKEY_EN
GSM_RESET
USB_VBUS
GSM_PWR
VDD_EXT
GND
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File: Mega2560_Custom_v2.0.PrjPcb
Date: 5/11/2023
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Drawn By: Huzaifa Ali
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ATMEGA2560
Analog Inputs
J11
HA0200500000G
U12
ATMEGA2560-16AU
U14
TLV9301IDCKR
U15
TLV9301IDCKR
Fc = 106Hz
Fc = 106Hz
J12
BH254V-10P (JTAG 2.54mm)
J13
ICSP
X1
X322516MLB4SI
SW1
FB10
LED10
OLED Module
LED7
DNP
LED6
Yellow
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USB-to-UART for Programming

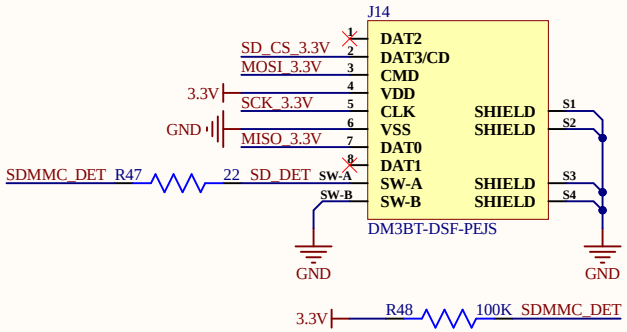


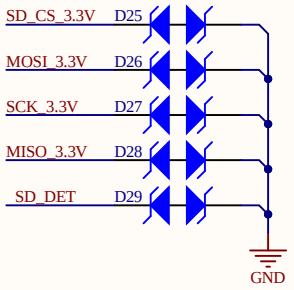


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SD Card Interface











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